

Process Characterization Plan

**A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) —
PCP-008**

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Virology laboratory	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; Cpk, one-sided

process capability index; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; LOQ, limit of quantitation; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PRESS, predicted residual error sum of squares; qPCR, quantitative polymerase chain reaction; RSM, response surface methodology; SOP, standard operating procedure; TCID₅₀, fifty percent tissue culture infectious dose; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Anion exchange chromatography is the final chromatographic step of the A-Mab drug substance process, and it is operated in the flow-through mode. This plan defines the process characterization study for that step, and it is written before the study is executed, so it states what will be done and how the result will be judged, and it states no findings.

The study provides the Stage 1 process design record for the anion exchange step under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and it supports the transfer of the drug substance process from the development site to the commercial site (PTP-001). The ranges justified here are the ranges that will be carried into process performance qualification.

The plan covers the 5 process parameters of the step (Table 6.1), the quality attributes the step governs (Table 4.1), and the qualified scale-down model on which every study run will be performed. The 4 measured responses are pool host cell protein, the log reduction factors for XMuLV and for MVM, and step yield. The work will be executed at development scale and analysed by design of experiments, in the enhanced development approach described in ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012).

The column itself is outside the scope of this plan. Packing, sanitization and resin life-cycle qualification are covered by SOP-2008 and are not characterized here. Confirmation of the operating ranges at commercial scale belongs to Stage 2 and to the process performance qualification, and the cumulative viral clearance claim for the drug substance is consolidated in PCMR-001, so this plan covers only the contribution the anion exchange step makes to that claim.

2 Objectives

The study is designed to meet the five objectives listed below, and PCR-008 will report the outcome against each of them in turn.

1. Quantify the effect of load pH, equilibration and wash-1 conductivity, load conductivity and protein load on pool host cell protein, on XMuLV and MVM clearance and on step yield, over the ranges given in Table 6.1.
2. Define a multivariate operating region in the parameters studied multivariately, within which every governed quality attribute meets its acceptance criterion.
3. Establish a proven acceptable range for each parameter and each governed attribute, and confirm that the normal operating ranges lie inside those ranges.
4. Classify every parameter of the step under SOP-4001, from the effects the study demonstrates and from the risk of operating outside the region.
5. Establish the log reduction the step contributes to the cumulative MVM and XMuLV claims, so that the modular clearance argument can be consolidated in PCMR-001 (International Council for Harmonisation 2023a).

3 Related documents

The parent documents and the paired report are listed in Table 3.1. RA-001 sets the scope of this study, PCMP-001 sets the approach common to every characterization plan in the campaign, and PCR-008 is the report in which the executed study will be described.

Table 3.1: Related documents in the A-Mab characterization package.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-008	Process Characterization Report (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and the validated analytical methods this study depends on are listed in Table 3.2.

Table 3.2: Controlled procedures and validated analytical methods for the anion exchange characterization study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Reference	Title	Type
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuvLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The anion exchange step is a flow-through polish. The column is packed with a strong anion exchange resin and equilibrated at a pH that lies above the isoelectric point of the acidic process impurities and below that of the antibody. Under those conditions A-Mab carries a net positive charge and passes through the column. Host cell protein, DNA, endotoxin and virus particles bind to the resin instead. Product is collected in the flow-through and in the post-load wash, so the step purifies without a bind and elute cycle, which is why its yield is high and why its capacity is set by the impurity burden and not by the product.

Load pH and conductivity govern the separation. Load pH sets the charge on the antibody and on the impurities, and conductivity sets how strongly the buffer counterions compete with those impurities for the resin surface. As load pH falls the antibody becomes more positively charged and the acidic impurities less negatively charged, so binding weakens and clearance falls. As conductivity rises the same binding is screened and clearance falls again, and the equilibration and wash-1 buffer acts on this equilibrium at the point where a weakly bound impurity is most likely to be displaced into the pool, which is why its conductivity is treated as a parameter in its own right. Protein load sets the mass challenge on the column, and at high load the impurity burden approaches the binding capacity available to hold it.

The platform is well established. A flow-through anion exchange polish of this design has been used to manufacture three prior products (X-Mab, Y-Mab and Z-Mab), and the mechanism of impurity and virus removal is the same for A-Mab as it is for those products (CMC Biotech Working Group 2009). Prior knowledge therefore sets the expectations that this study will confirm or refute, but it does not by itself define an operating region, because the ranges, the interactions and the position of the acceptance limits are specific to this product and this load material.

The mechanistic expectation is specific enough to be tested. Clearance of both model viruses should fall as load pH falls and as load conductivity rises, and the two factors should interact, because both act on the same electrostatic equilibrium. Pool host cell protein should rise with protein load, with equilibration and wash-1 conductivity, and as load pH falls. Step yield should be high and largely insensitive across the ranges studied, since the product does not bind to the resin.

4.2 Quality attributes in scope

The quality attributes this step governs are listed in Table 4.1, and the two viral safety attributes carry the highest criticality in that set.

Table 4.1: Quality attributes governed by the anion exchange step, with acceptance criteria and criticality from the drug substance register.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Leached Protein A	process impurity	0–5 ppm	L-M	16
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

The cumulative clearance of MVM is assigned to this step in the drug substance CQA register. MVM is a non-enveloped parvovirus, so it is not inactivated by the low-pH hold at Step 6, and the claim rests on the anion exchange step and the small virus retentive filter together. Those two steps remove virus by different mechanisms, which is what makes their contributions additive under ICH Q5A(R2) (International Council for Harmonisation 2023a). The step is credited with XMuLV clearance as well, and there the low-pH hold (PCP-006) carries the larger share while the filtration step (PCP-009) carries a further increment.

Host cell protein is the impurity attribute that drives the design of this study, and its criticality is moderate to high (Table 4.1). Three steps of the train remove it. Protein A capture (PCP-005) removes the bulk of the burden, cation exchange (PCP-007) removes most of the remainder, and this step provides the final reduction before formulation. Residual DNA and leached Protein A are also reduced here, and the register assigns residual DNA a very low criticality and leached Protein A a low to moderate one (Table 4.1), so neither is expected to constrain the operating region. Both will be measured at every design point so that the expectation can be checked.

No claim is made for aggregate removal across the anion exchange step, since aggregate is polished at cation exchange (PCP-007). The charge variant and glycan distributions are formed upstream and are not modified by the purification steps themselves (CMC Biotech Working Group 2009), so they are not responses of this study.

4.3 Risk-based prioritization of parameters

The pre-characterization risk assessment RA-001 decided which parameters are studied and how, and its output for this step is reproduced in Table 4.2. The method is risk ranking and filtering. A severity score for the potential impact on a quality attribute is combined with the likelihood of the failure mode and with the ability to detect it, and the resulting score carries the parameter into a multivariate design, into a univariate study, or into neither (International Council for Harmonisation 2023b). Criticality is treated as a continuum and not as a binary classification (CMC Biotech Working Group 2009).

Table 4.2: Pre-characterization risk assessment for the anion exchange step, from RA-001. The assigned study type is the scope this plan inherits.

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	RPN	Assigned study	Pri- or- ity
Load pH	Load pH below characterized range (<7.2)	Reduced HCP removal and reduced XMuLV/MVM clearance	10	7	10	700	multi-variate DoE	High
Equil/Wash 1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Reduced HCP removal	6	7	8	336	multi-variate DoE	Medium
Load conductivity	Load conductivity above characterized range	Reduced viral clearance (interacts with pH)	10	7	10	700	multi-variate DoE	High
Protein load	Protein load above characterized range (>300 g/L resin)	Reduced HCP removal capacity (viral clearance robust)	6	7	8	336	multi-variate DoE	Medium
Operating flow rate	Operating flow rate above 450 cm/hr	Reduced residence time / viral clearance margin	10	7	10	700	univariate (justified)	High

Load pH, load conductivity and operating flow rate carry the highest score in Table 4.2. Load pH and load conductivity score 700, because a low load pH reduces both host cell protein removal and viral clearance, and a high load conductivity reduces clearance in the same way. A loss of viral clearance is scored at the top of the severity scale. Operating flow rate carries the same score for the same attribute, but RA-001 assigned it a univariate study, because it acts on the separation through residence time and

not through the buffer chemistry that sets the binding equilibrium. Equilibration and wash-1 conductivity and protein load score 336, since their expected effect is on host cell protein rather than on viral safety.

The 4 parameters assigned to a multivariate study are the factors of the designed experiments described below, and flow rate is assessed one factor at a time. No parameter of this step was left unstudied, so the study scope is the whole parameter set of the unit operation and not a filtered subset of it.

5 Materials and methods

5.1 Scale-down model and its qualification

Every run in this study will be performed on a qualified scale-down model of the commercial column, and that qualification is what allows the results to be applied to commercial manufacture (SOP-1001).

The model is designed on the established principles of chromatography scaling. Bed height, linear velocity, protein load per litre of resin, buffer composition and the volume of each phase normalized to column volumes are held equal to the commercial values. Column efficiency is measured before use by plate count and peak asymmetry. A column that fails the packing criteria of SOP-2008 is repacked before any study run. Residence time at the model scale therefore matches residence time at commercial scale for the same linear velocity, which is the property a flow-through separation depends on.

Qualification will compare the model against commercial scale data at set-point conditions, and replicate runs at the set-point will be compared with the at-scale record for step yield, pool host cell protein and pool volume. The model will be accepted when the means do not differ significantly and when every model result falls inside the range observed at scale. The comparison is documented under SOP-1001. Its completion is a precondition for starting any characterization run.

Viral clearance cannot be measured at manufacturing scale, because live virus may not be introduced into a manufacturing facility. The clearance runs will therefore be executed in a dedicated biosafety laboratory, on a separately qualified small-scale spiking model, following ICH Q5A(R2) (International Council for Harmonisation 2023a). The virus spike is added to the load at a small volume fraction, so that the composition of the load is essentially unchanged, and the spiked load is assayed alongside the pool so that the log reduction factor is the difference between two measurements from the same run. The qualification of the spiking model, and not the size of the column, is what supports a clearance claim.

Centre point runs are replicated so that the pure error of the system can be estimated independently of the fitted model. The screening design carries 3 centre points and the response-surface design carries 4, which give 2 and 3 degrees of freedom for pure error. Those terms are the reference against which lack of fit will be tested.

5.2 Operation

The step will be operated as it is operated at commercial scale, under SOP-2011. The column is equilibrated with the wash-1 buffer, the load is applied at the target protein load and linear velocity, the column is washed with the same buffer, and the product is collected across the flow-through and the wash. Pool collection starts and stops on ultraviolet absorbance thresholds, and the pool is held refrigerated until it is sampled.

The load material is the cation exchange pool from Step 7. A single pool, or a set of pools blended to one specification, will be used for the whole study, so that load composition is not a hidden factor in the design. Before the study starts the load will be confirmed representative. Protein concentration, pH and conductivity will be adjusted to the study target, and the acidic charge variant content will be measured by icIEF (AMV-3013) and confirmed to lie within the range seen in routine cation exchange pools. The hold time between neutralization of the cation exchange pool and its use as anion exchange load will be kept within the limit recorded in SOP-2011, because an extended neutral hold raises the deamidated fraction of the load and changes the charge distribution that this step has to separate.

Buffers, resin and the column are inputs controlled to platform specification. They are identity and quality controlled under SOP-2011 and SOP-2008. They are not parameters of this study, and they are held at their platform values throughout it. Resin lot and column use history will likewise be held constant across the study, so that the fitted effects cannot be confounded with resin age.

5.3 Analytical methods

Every response is measured by a validated method, and the validated performance of the methods that carry the conclusions of this study is given in Table 5.1.

Table 5.1: Validated performance of the analytical methods that measure the study responses.

Method Title		Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3012	Host-Cell Protein by ELISA	9.5 %RSD	1 ng/mg	95	0.4
AMV-3016	Leached Protein A by ELISA	6.5 %RSD	0.25 ppm	97.4	0.41
AMV-3017	XMuvLV Infectivity Titre (TCID50)	0.3 log10 SD	1.5 log10 TCID50/mL	96	0.28
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	0.32 log10 SD	1.4 log10 TCID50/mL	96	0.3

Pool host cell protein is measured by ELISA (AMV-3012). Its intermediate precision is 9.5% relative standard deviation, against 6.5% for the leached Protein A ELISA (AMV-3016), and those are the only two methods in Table 5.1 whose precision is reported on a relative scale. A difference in pool host cell protein therefore has to stand clear of a relative standard deviation of that size before it can be attributed to a factor of the design.

XMuvLV titre is measured by an infectivity assay (AMV-3017) and MVM titre by infectivity with a molecular confirmation (AMV-3018), while residual DNA is measured by qPCR (AMV-3014). The precision of the two infectivity assays is reported as a standard deviation on the log10 scale, which is not comparable with a relative standard deviation. Both carry an intermediate precision of about 0.32 log10 (Table 5.1), so a difference of that order between two clearance runs sits inside the noise of the assay and will not be interpreted as an effect of a process parameter.

Step yield is calculated from the product mass in the load and in the pool, each of them the product of a measured protein concentration and a measured volume, and protein concentration is measured by ultraviolet absorbance under AMV-3019. Validated performance for that method, for the charge variant method (AMV-3013) and for the residual DNA method (AMV-3014) is recorded in the respective method validation reports and is not reproduced here.

Where the virus titre of a pool falls below the quantitation limit of the infectivity assay (Table 5.1), the log reduction factor is a lower bound and will be reported as such. A lower bound is used as it stands in the modular clearance summary, and it is never converted into a point estimate.

5.4 Sampling plan

Samples will be taken from the same points in every run, so that the study is balanced in the analytical dimension as well as in the factors.

Each run of the non-viral designs is sampled at the load and at the pool. The load is sampled before application for protein concentration, pH, conductivity, host cell protein and charge variants, and the flow-through pool is sampled after collection for protein concentration, host cell protein, residual DNA and leached Protein A. A retain of each pool is stored frozen, so that an unexpected result can be re-tested without repeating the run.

The viral runs are sampled differently. The spiked load and the flow-through pool are assayed for infectious virus, and a cytotoxicity and interference control is run for each matrix, as ICH Q5A(R2) requires (International Council for Harmonisation 2023a).

Centre point runs are sampled exactly as the factorial runs are. They will be executed at intervals through the study and not consecutively, so that the pure error term reflects the variability of the whole campaign and not the variability of a single day.

5.5 Statistical methods

The analysis follows SOP-1002. Each factor is coded so that the midpoint of its characterization range is zero and the two edges are ± 1 , and every model is fitted to the coded factors, so that estimated effects are comparable across factors carrying different units. Centre point runs sit at the midpoint of every characterization range, which is the common reference condition of both designs.

The screening study will be analysed with a model carrying every main effect and every two-factor interaction, fitted by ordinary least squares. An effect is reported as twice the coded coefficient, which is the change in the response between the two edges of the studied range. Terms are judged at $\alpha = 0.05$, and the screening study identifies which factors and which interactions act on each response. It is not the predictive model, and no operating range will be defined from it.

The response-surface study will be analysed with the full quadratic model in the 4 multivariate factors. That model carries 15 terms against 28 runs and leaves 13 residual degrees of freedom, and it is the predictive model, so it is the basis of the operating region, of every proven acceptable range and of the capability estimates in this plan. Model adequacy will be judged on four statements taken together. These are the significance of the model F test, the coefficient of determination and its adjusted form, the predicted coefficient of determination computed from the PRESS residuals, and the lack of fit test against pure error. Residuals will be inspected against the predicted values, against the run order and on a normal probability plot.

A factor with no significant effect on any response is a result and not a gap. It will be classified on the evidence of that absence. It will be retained in the knowledge space as evidence that the governed attributes are robust to it across the range screened.

Commercial scale capability will be estimated by Monte-Carlo simulation of 2,000 batches, with every parameter drawn inside its normal operating range and the fitted models used to predict each attribute. Capability will be reported as a one-sided Cpk against the relevant limit. This plan sets no numerical capability threshold. The report will state each Cpk with the margin between the predicted distribution and the limit, and the acceptability of that margin will be argued from the criticality of the attribute and the control capability of the parameters.

6 Study design

The design has to separate 4 factors that act on one equilibrium, so the multivariate work is done in two stages. A screening study identifies the active factors and interactions, a response-surface study models the region they define, and operating flow rate is assessed one factor at a time for the mechanistic reason given below.

6.1 Factors, ranges and study type

The parameters, their set-points, the ranges to be studied and the normal operating ranges are given in Table 6.1, and the coded letters used for the 4 multivariate factors in the design matrices are given in Table 6.2.

Table 6.1: Parameters of the anion exchange step, with the ranges to be studied, the normal operating ranges and the assigned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Load pH	pH	7.5	7.2–7.8	7.35–7.65	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	1.6–3.6	2.1–3.1	multivariate
Load conductivity	mS/cm	5.5	3–8	4–7	multivariate
Protein load	g/L resin	200	50–300	120–280	multivariate
Operating flow rate	cm/hr	300	150–450	200–400	univariate

Table 6.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

Every studied range in Table 6.1 is wider than the corresponding normal operating range, and deliberately so, because a characterization range has to bound the knowledge space, which means it has to include settings the commercial process will not

normally see (CMC Biotech Working Group 2009). The wider ranges also support later movement within the operating region without new studies, and each range was set from prior knowledge and from the mechanism.

Load pH will be studied over 7.2–7.8 against a set-point of 7.5 (Table 6.1). The lower edge sits below the pH at which prior products began to lose virus binding, so that the loss of clearance is bounded by measurement instead of by extrapolation. The upper edge is limited by antibody stability at the load protein concentration.

Equilibration and wash-1 conductivity will be studied over 1.6–3.6 mS/cm, where the lower edge is a more dilute buffer than the commercial site will prepare and the upper edge lies above the conductivity at which platform experience shows host cell protein beginning to appear in the flow-through.

Load conductivity will be studied over 3–8 mS/cm. The load enters this step from the cation exchange pool, and its conductivity is the input property most likely to move from batch to batch, so the range is wide enough to cover a cation exchange pool taken from either end of its own operating region.

Protein load will be studied over 50–300 g/L resin, and the upper edge lies above the highest load the commercial column will be asked to carry, so that the capacity of the step is bounded by data. The lower edge lies well below the set-point, because a response-surface model may only be used inside the region it was fitted over, and the low end of the load range has to be part of that region.

Operating flow rate will be studied over 150–450 cm/hr, one factor at a time.

6.2 Screening design

The screening study is a two-level full factorial in the 4 multivariate factors, augmented with 3 centre points, giving 19 runs in total. The planned matrix is given in Appendix A. A full factorial was chosen in preference to a fractional design because a design in 4 factors is small enough to afford it, and because the expected interactions are between factors that act on the same equilibrium. No interaction may be aliased with a main effect at this step, since an interaction between load pH and load conductivity is the effect the viral clearance claim is most sensitive to.

The design estimates all 4 main effects and all 6 two-factor interactions independently, and it leaves 8 residual degrees of freedom for the two-factor interaction model. Runs will be executed in a randomized order, with the centre points distributed through the sequence. The purpose of this study is to identify which factors and interactions act on each response, and the magnitudes it estimates will be refined by the response-surface study.

6.3 Response-surface design

The response-surface study is a face-centred central composite design in the same 4 factors, and it carries 16 factorial runs, 8 axial runs and 4 centre points, giving 28 runs in total. The planned matrix is given in Appendix B. The axial points sit on the faces of the cube and not outside it, so no run is executed beyond the characterization ranges in Table 6.1. That matters here, because those ranges are the edges of the knowledge space and the study is not permitted to operate the model outside them.

All 4 multivariate factors are carried into the response-surface design. None can be excluded in advance, since each is expected to act on at least one of the 4 responses and the two conductivities are expected to interact with load pH. The design adds the curvature terms that a two-level screening design cannot estimate. The fitted quadratic model is the model from which the operating region, the proven acceptable ranges and the capability estimates will be derived. It is the only model in this study that will be used to predict.

6.4 Univariate assessment

Operating flow rate will be assessed one factor at a time. It will be run at its set-point and at each edge of 150–450 cm/hr, with the 4 multivariate factors held at their set-points, and the same responses will be measured at each setting.

The reason it is held out of the multivariate design is mechanistic. Flow rate acts on the separation through residence time, and residence time governs how closely impurity binding approaches equilibrium. It does not change the charge state of the antibody or of the impurities, which is how the multivariate factors act, so an interaction between flow rate and the solution properties is not expected. RA-001 assigned it a univariate study on that basis (Table 4.2). If the univariate runs show an effect on viral clearance at the high edge of the range, the parameter will be carried into a further multivariate study and this plan revised under change control.

7 Acceptance and decision criteria

This section states the criteria the study must meet, before any data exist. They apply to the scale-down model, to the fitted models and to the measured quality attributes.

The scale-down model must be qualified before any characterization run starts. The criterion is the one given above under materials and methods. There must be no significant difference between the model and the commercial scale record on step yield, pool host cell protein and pool volume, and every model result must fall inside the range observed at scale (SOP-1001). Runs performed on a model that has not met that criterion will not be used to define an operating range.

A fitted model will be used to define the operating region only if it is adequate. The model F test must be significant at $\alpha = 0.05$, the lack of fit test against pure error must not be significant at the same level, and the predicted coefficient of determination must be close enough to the adjusted value to show that the model predicts and does not merely fit. A significant lack of fit will be investigated before the model is used, and both the investigation and its outcome will be reported in PCR-008. Where a response yields no significant model, that will be reported as an absence of effect over the ranges studied, and the parameters will be classified on that evidence.

The acceptance criteria for the measured responses are given in Table 7.1, together with the basis of each.

Table 7.1: Acceptance criteria for the measured responses, with the basis of each.

Re- sponse	Quality attribute	Accep- tance criterion	Unit	Basis
Pool HCP (ng/mg)	Host Cell Protein (HCP)	≤ 100	ng/mg	Drug-substance specification
XMULV LRF (log ₁₀)	Viral clearance — XMULV (enveloped)	≥ 3.78	log ₁₀ (this step)	Cumulative requirement less the clearance credited to the other steps
MVM LRF (log ₁₀)	Viral clearance — MVM (parvovirus)	≥ 3.28	log ₁₀ (this step)	Cumulative requirement less the clearance credited to the other steps

Pool host cell protein is judged against the drug substance specification, which is a ceiling of 100 ng/mg (Table 7.1). The two viral clearance responses are judged against a step-level requirement instead, because the specification is cumulative across the

process and no single step is asked to meet it alone. The cumulative MVM requirement is 8.6 log₁₀ (Table 4.1), and subtracting the clearance credited to the other steps of the train leaves 3.28 log₁₀ for this step, while the same calculation gives 3.78 log₁₀ for XMuLV. Those two figures are what the operating region has to satisfy here, and step yield carries no acceptance criterion at all, since it is measured as a process performance response.

The operating region will be declared acceptable when the fitted response-surface models predict every governed attribute inside its criterion across the whole region, and when the normal operating ranges in Table 6.1 lie inside it. The worst case corner of the region follows from the mechanism, and it is the corner at the lowest load pH, the highest load conductivity, the highest equilibration and wash-1 conductivity and the highest protein load. Clearance is expected to be lowest there and pool host cell protein highest, so the region will not be declared acceptable unless the prediction at that corner meets both criteria.

If a governed attribute fails inside the characterized ranges, the region will be reduced to the sub-region that meets the criteria, and the boundary will then be reported as an edge of failure. If that boundary falls inside a normal operating range, the normal operating range itself will be revised, and the revision will be handled under the pharmaceutical quality system (International Council for Harmonisation 2008).

Departures from this plan will be recorded and impact-assessed in PCR-008. A departure that leaves a run inside its designed setting will be assessed and the run retained. A departure that makes a run unrepresentative of the commercial process invalidates the affected design, and a load material outside the acceptance stated above under operation is such a departure. In that case the affected runs will be repeated in full on requalified material, and the invalidated data will be retained for the root cause investigation and excluded from the reported analysis.

8 Proven acceptable ranges (planned analysis)

The operating region is multivariate and a proven acceptable range is not. Both will be reported, and this section states how the ranges will be derived and what they will mean.

The acceptance basis is the one given in Table 7.1. For pool host cell protein the criterion is the drug substance specification. For the two viral clearance responses it is the step-level requirement back-calculated from the cumulative requirement, and no other value is used in this analysis.

Two analyses will be run for each governed attribute and each response-surface factor. The first holds the other parameters at their set-points and evaluates the fitted response-surface model across the parameter's characterization range. The second varies the other parameters inside their normal operating ranges, drawing 2,000 Monte-Carlo samples of those parameters at each of 81 points across the range, predicting the attribute from the fitted model with its residual variance included, and taking the 95% predictive interval of the result.

The decision rule is the same in both analyses. The proven acceptable range is the contiguous interval of the parameter, containing the set-point, over which the attribute meets its criterion. In the first analysis that test is applied to the predicted mean, and in the second it is applied to the whole predictive interval. The second rule is the stricter of the two, so where the two ranges differ the narrower one is the range the control strategy will rely on.

Each governed attribute will be reported with a plot of its governing parameter, which is the parameter carrying the largest main effect on that attribute. The plot carries the parameter on the horizontal axis and the attribute on the vertical axis, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked. The two analyses appear as the two panels of the same figure.

The relationship between a proven acceptable range and the operating region has to be stated, because the two are easily confused. A proven acceptable range is a statement about one parameter while the others are held or varied in a defined way. The operating region is a statement about all 4 factors together. Where a whole characterization range is proven acceptable for every governed attribute, the binding constraint on the process is the multivariate corner of the region and not any univariate range, and the report will say so explicitly.

Every range reported by this method is bounded by the characterization ranges in Table 6.1 and by the fitted model. It says nothing about settings outside those ranges,

and nothing about a load material that falls outside the acceptance stated above under operation.

9 Data management and integrity

Study records will be managed to ALCOA+ expectations. Chromatography runs are recorded by the instrument control system and analytical results by the laboratory information management system, and both records are retained with their audit trails for the review that closes the study. The statistical analysis will be performed in the validated environment described in SOP-1002, and the analysis scripts and the data extracts are retained with the study record.

Any value transcribed by hand will be verified by a second person against the source record. Data excluded from an analysis will be identified in PCR-008 with the reason for its exclusion, and no result will be removed from the record itself. Raw data, analysis files and the report are retained for the life of the product.

10 Roles and responsibilities

The functions that carry work under this plan are listed in Table 10.1, and each of them appears in the approval record at the front of the document as author, reviewer or approver.

Table 10.1: Functions and responsibilities under this plan.

Function	Responsibility under this plan
Process Development / MSAT	Owns this plan, executes the non-viral runs, performs the statistical analysis and writes PCR-008.
Analytical Development	Releases the validated methods, analyses the samples and provides the method performance record.
Virology laboratory	Executes the spiking runs under ICH Q5A(R2) and reports the titres and log reduction factors.
Manufacturing Science	Provides the commercial scale record used to qualify the scale-down model.
Statistics	Reviews the designs before execution and the fitted models before they are used.
Quality Assurance	Approves this plan, reviews every departure from it and approves PCR-008.

The responsibilities in Table 10.1 are assigned to functions and not to individuals, so the plan does not need revision when staff change.

11 Deliverables and schedule

The deliverable of this plan is PCR-008, the process characterization report for the anion exchange step. It reports the executed designs, the fitted models, the operating region, the proven acceptable ranges, the estimated capability and the classification of every parameter, and it records every departure from this plan.

The study runs as a sequence of dependent phases, and the scale-down model is qualified first. The screening study runs next, and its analysis is completed before the response-surface study starts, so that the design of the second study can be confirmed before further runs are committed. The response-surface study follows, while the viral clearance runs are executed in parallel in the biosafety laboratory, on the same design points, because they use a different facility and a separate qualification. The univariate flow rate runs may be executed at any point after the scale-down model is qualified, and analysis and reporting complete the study.

No calendar dates are given in this plan, since the sequence and the dependencies above are the schedule, and the executed dates will be recorded in PCR-008.

12 Risks and assumptions

The principal risk to this study is that the scale-down model does not represent the commercial column, and it is managed by the qualification described under materials and methods, which is the reason no characterization run may start before that qualification is complete. A residual risk remains after qualification, because the comparison is made at set-point conditions and representativeness at the edges of the ranges is inferred from the scaling principles rather than demonstrated. Ranges defined on the model are not confirmed at commercial scale until Stage 2, and PCR-008 will state that bound on every range it reports.

The virus spiking model carries assumptions of its own. The model viruses are laboratory strains chosen as representatives of a class, the spike is added to a load that is not identical to the commercial load, and the clearance measured is a property of that model system (International Council for Harmonisation 2023a). These assumptions are accepted because the same convention is applied at every clearance step of this process and was applied to the three prior products, so the modular claim is at least internally consistent.

Load material supply is a practical risk. The study needs one representative cation exchange pool, or a set of pools blended to a single specification, and the quantity required is large in comparison with a development batch. If the load has to be prepared from more than one cation exchange run, the pools will be blended before any study run and the blend characterized as described under operation.

The study assumes that the platform mechanism transfers from the three prior products to A-Mab, which is what makes this a confirmatory study instead of an exploratory one. It also assumes that the resin lot used in the study represents the resin the commercial process will use, and that assumption is managed by the resin qualification in SOP-2008. Both are stated here so that the report can test them, and if either fails the affected conclusions will be re-derived and not defended.

13 Approvals

This plan takes effect when it has been approved by each function named in the approval record at the front of the document. Changes after approval are made by revising the plan under change control (International Council for Harmonisation 2008). A departure identified during execution is not a plan change. It is recorded and assessed in PCR-008, as described above under acceptance and decision criteria.

14 Appendices A and B — Planned design matrices

The two appendices give the designs as they will be executed. Neither carries a response column, because the study has not been performed.

14.1 Appendix A — Planned screening design matrix

The planned screening design is given in Table 14.1 in coded units, with the type of each run. The factor letters are defined in Table 6.2.

Table 14.1: Planned screening design matrix in coded units.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

14.2 Appendix B — Planned response-surface design matrix

The planned response-surface design is given in Table 14.2 in coded units. Run order will be randomized at execution and the executed order recorded in PCR-008.

Table 14.2: Planned response-surface design matrix in coded units.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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